

From spectral change to a beat pulse

Tempo candidates and the causal PLP tracker implemented in sllr.

05.01 · Periodicity suggests tempo

EXPLANATION

The offline tempo plot autocorrelates the spectral-flux sequence. Repeated attacks reinforce one another at particular delays; converting those delays to beats per minute gives candidate tempos.

The search covers 40–240 BPM. Peaks can reflect subdivisions or every-other-beat patterns, so half- and double-tempo interpretations remain possible. A short or nonperiodic recording may have no useful tempo.

Autocorrelation scores describe repetition. They are not probabilities that a particular tempo is correct.

MATHS

For nonnegative flux Δ_m and lag ℓ ,

$$R[\ell] = \frac{\sum_{m=0}^{M-\ell-1} \Delta_m \Delta_{m+\ell}}{\sum_{m=0}^{M-1} \Delta_m^2}.$$

The conversion from lag to tempo is

$$\text{BPM}(\ell) = \frac{60f_s}{H\ell}.$$

The denominator uses total activation energy rather than a lag-dependent normalization. Negligible energy yields no tempo candidates.

05.02 · Past onsets vote for a pulse

EXPLANATION

The causal PLP tracker evaluates rhythmic periodicities from recent activation history. It chooses a tempo and phase, constructs a signed rhythmic kernel, and overlap-adds successive kernels into a pulse buffer.

slr uses spectral-flux activation rather than a trained onset model. The default tempo grid is 60–180 BPM in increments of two; its kernel setting is six seconds. These are distinct from the offline autocorrelation search.

The live audio front end uses trailing windows. Future portions of a pulse kernel are predictions from past information, not measurements of future audio.

MATHS

Let δ be the activation-frame period and $h = \max(2, \text{round}(T_k/(2\delta)))$. For lag $j = 0, \dots, h$,

$$w_j = \frac{1}{2} + \frac{1}{2} \cos(\pi j/h).$$

For tempo b on the grid, the implementation computes

$$z_b = \sum_{j=0}^h \Delta_{n-j} w_j e^{i2\pi(b/60)\delta j}.$$

It chooses $b_* = \arg \max_b |z_b|^2$ and $\phi = \arg z_{b_*}$, then adds the centered kernel

$$\kappa_n[j] = \frac{w_{|j|}}{h} \cos\left(2\pi \frac{b_*}{60} \delta j + \phi\right), \quad -h \leq j \leq h.$$

The pulse buffer shifts one step before each new contribution.

05.03 · Beat decisions and timing limits

EXPLANATION

A beat event requires a local pulse maximum, sufficient clipped pulse strength, non-negligible activation, a warm-up period, and enough time since the previous event. The default lookahead is zero and the threshold is 0.08.

The quantity named stability in the implementation is a clipped pulse sample, not a calibrated confidence probability. Increasing lookahead samples a predicted future pulse position.

Microphone capture, frame buffering, and display all add latency. The implementation does not establish zero device latency or the accuracy of a trained system. Weak onsets and tempo ambiguity remain practical limits.

MATHS

For lookahead L_a , the decision index is

$$d = \text{clamp}(\text{round}(h + L_a/\delta), 1, 2h - 1).$$

If Γ is the overlap-added pulse, stability is $s = \text{clamp}(\Gamma[d], 0, 1)$. A local maximum must satisfy

$$\Gamma[d] > \Gamma[d - 1], \quad \Gamma[d] \geq \Gamma[d + 1].$$

The elapsed-time conditions are

$$t \geq \min(T_k/2, 2), \quad t - t_{\text{last}} > \frac{24}{\max(1, b_*)}.$$

The refractory duration is 40% of the selected beat period. These conditions describe the current code, not a universal beat-tracking definition.